

ABHISHEK V S

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Bengaluru, Karnataka - 563123, India

PROFESSIONAL SUMMARY

Experienced Data Scientist specializing in deploying machine learning solutions, with a focus on Text-to-Speech (TTS) technologies. Proficient in advanced AI methodologies, including RAG, Langchain, LLMs, and Prompt Engineering. Proven track record in enhancing TTS systems by optimizing response quality and processing speed, resulting in notable improvements in naturalness and intelligibility. Eager to leverage my expertise in a challenging data science role to drive innovation and deliver impactful results.

EXPERIENCE

• Celebal Technologies

Associate Data Scientist

April 2023 - Present

Hydrabad, India

- Developed and deployed a sophisticated QnA Chatbot using LLMs and RAG.
- Added real-time trading data and streaming, enhancing user engagement by 35%.
- Enhanced data ingestion latency for PDF data by through asynchronous processing.
- Implemented hybrid search improving the accuracy of the bot.
- Integrated real-time API data for dynamic updates and employed advanced prompt techniques such as Chain of Thought (CoT) Prompting, improving response accuracy and relevance by 12%.
- Implemented performance optimization strategies that increased TTS output quality, achieving a 20% improvement in naturalness and intelligibility metrics.

• Flip Robo Technology

Data Science Intern

March 2022 - September 2022

Bangalore, India

- Developed machine learning models for regression and classification.
- Performed exploratory data analysis to extract insights and identify relationships within the data.
- Visualized data by distinguishing between continuous and categorical variables.

PROFESSIONAL PROJECTS

• GenAI Legal Bot for Adani Cements

Tools: Azure AI Search, Python, Flask, App Services, OpenAI, Azure Storage, Azure NoSQL

Nov 2023 - Jan 2024

- Led the development of a GenAI-based Legal Bot tailored for lawyers and admins at Adani Cements.
- Enabled the bot to process and interact with a wide range of document formats, including PDF, JPG/PNG, Excel, CSV, and DOCX.
- Implemented a Hybrid AI search system, leveraging Azure AI Search, for efficient document retrieval and integrated OpenAI LLMs for dynamic and accurate query generation.
- Developed a feature that allows users to query and extract information from both structured and unstructured data within documents.
- Incorporated side navigation functionality that highlights the relevant text from documents and shows page numbers, allowing users to view and interact with the exact document excerpt.
- Designed a summarization feature for lawyers, enabling them to generate concise summaries of lengthy documents by clicking a designated button.
- Empowered admins to add direct instructions to the bot for customized responses and enhanced functionality.

• GenAI Bot for Dubai Airports (POC)

Tools: Python, FastAPI, Azure OpenAI, Azure Services, Flask, Azure App Services, GIT

Feb 2024 - Apr 2024

- Developed a GenAI chatbot for Dubai Airports to assist both passengers and corporate users with their inquiries and services.
- Enabled passengers to easily check flight status, navigate the airport, and access transportation services, improving their overall travel experience.
- Integrated real-time data for flight updates, navigation, and transportation, ensuring dynamic and accurate responses for passengers.
- Designed and implemented a dual-purpose interface allowing users to select between passenger services and corporate services.
- For corporate users, provided assistance with operational inquiries, flight schedules, and logistical support.
- Applied advanced prompt engineering and integrated Azure OpenAI for intelligent, context-aware responses, ensuring a smooth interaction for both passengers and staff.

- **GenAI ChatBot for ADSS Traders and Dashboard** May 2024 - July 2024
Tools: Python, FastAPI, Uvicorn, Azure OpenAI, Azure Services, VeraCode, GIT, API Management, Jira, Confluence
 - Led development of an advanced GenAI chatbot with OpenAI LLMs.
 - Integrated real-time API data for dynamic updates and applied advanced prompt engineering for optimized knowledge retrieval.
 - Integrated deep link functionality in chatbot responses for seamless user navigation and interaction.
 - Designed and implemented real-time streaming response system for the chatbot.
 - Developed a dashboard to monitor user engagement, track bot uptime and downtime, analyze RAGAS metrics for each query, and provide live conversation logs.
- **Bhashini Project - National Language Translation Mission** Sep 2024 - Present
Tools: Python, Docker, PyTorch, Virtual Machines, Azure ML
 - Lead research and development efforts for Text-to-Speech (TTS) models as part of Bhashini’s central government initiative under Digital India.
 - Fine-tuned two major open-source models: AI4Bharat and IITM, enhancing their performance for diverse language processing.
 - Conduct research on voice-cloning techniques to improve the adaptability and personalization of TTS services.

EDUCATION

- **Aditya College Of Engineering** July 2018 - July 2022
B Tech in Computer Science & Engineering Madanapalle, India
 - Grade: 70%
- **Swami Vivekananda PU College** June 2016 - July 2018
Pre-University College (PCMB) India
 - Grade: 70%
- **Sri Manjunatha English Medium School** June 2015 - June 2016
SSLC (Secondary Education) India
 - Grade: 73%

SKILLS

- **Programming Languages:** C, Python, R
- **Frameworks & Libraries:** Flask, FastAPI, Langchain, PyTorch 2.0, Pandas, Numpy, Scikit-learn, SpaCy, OpenCV
- **Data Science & Machine Learning:** Probability & Statistics, Web Scraping, Data Preprocessing, EDA, Classic ML, Deep Learning, Neural Nets, Speech Services, TTS Optimization
- **Natural Language Processing (NLP):** LLMs, Generative AI, Chatbots, Information Retrieval, Seq2Seq Modeling, Classical NLP
- **Cloud Technologies:** Microsoft Azure, Databricks
- **Tools & Technologies:** GIT, SQL, Docker, VeraCode, Jira, Confluence

CERTIFICATIONS

- **Machine Learning with Python - IBM** October 2023
- **PG Program in Data Science - DataTrained** October 2023
- **Big Data Certificate - DataTrained** October 2023
- **Tableau Certificate - DataTrained** October 2023
- **Azure AI Fundamentals - Microsoft** December 2023

ADDITIONAL INFORMATION

Languages: Kannada (Native), Telugu (Fluent), Hindi (Professional Working Proficiency), English (Professional Working Proficiency)
Interests: - Outdoor activities such as cricket and volleyball. - Keeping up with advancements in AI and machine learning. - Participating in hackathons and coding challenges. - Exploring emerging technologies and industry trends.
Professional Development: - Regular attendee of industry conferences and webinars. - Engaged in continuous learning through online courses and workshops.